Protein Sequence–Structure–Function Analytics and Predictive Modeling Using Deep Learning

Final Year Design Project Proposal
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1. Project Overview

This project aims to develop a deep-learning-based system for predicting the molecular functions of proteins from their amino-acid sequences and structure-derived information. Proteins perform essential biological activities, and predicting their functions can support biological research, drug discovery, and the interpretation of genetic variations.

The project will combine data analytics, deep learning, and software development. The research component will investigate how sequence and structural information can be used for protein function prediction. The software component will provide a Python-based backend, a MySQL database, and a web-based frontend for exploring proteins and viewing prediction results.

2. Problem Statement

A protein is a biological molecule made from a chain of amino acids. The order of amino acids is called the protein sequence. This chain folds into a three-dimensional structure, and that structure is closely related to the biological function of the protein.

Although many protein sequences are publicly available, the functions of many proteins remain unknown or incompletely annotated. Manual functional annotation requires expert biological knowledge and laboratory validation, making it time-consuming and expensive. Therefore, computational methods are needed to predict likely protein functions from available sequence and structural data.

This project will address the following research question:

Can deep-learning models predict the molecular functions of proteins using sequence and structure-derived information?

3. Objectives

- To collect and understand a publicly available protein sequence–structure–function dataset.
- To perform statistical analysis on protein sequences, structural contact information, and Gene Ontology function labels.
- To develop and train deep-learning models for protein molecular-function prediction.
- To compare sequence-based prediction with models that use both sequence and structural information.
- To develop a Python-based software system with a MySQL database and a web frontend.
- To evaluate model performance using appropriate multi-label classification metrics.

4. Dataset

The project will begin with the publicly available DeepFRI PDB–Gene Ontology benchmark dataset. This dataset was released by the authors of the research paper “Structure-based Protein Function Prediction Using Graph Convolutional Networks,” published in Nature Communications in 2021.

The dataset is derived from experimentally determined protein structures available in the Protein Data Bank (PDB). PDB is a public archive of three-dimensional structural data for biological molecules. The DeepFRI authors selected annotated protein chains from PDB and associated them with functional annotations.

The dataset contains protein sequences, structure-derived contact information, and Gene Ontology functional labels. Gene Ontology, commonly known as GO, provides standardized labels for protein functions. A protein may have more than one functional label; therefore, the prediction task is a multi-label

classification problem.

4.1 Data Sources

Source	Role in the dataset
PDB	Provides experimentally determined protein structures and associated chain-level sequence information.
SIFTS	Maps PDB protein chains to corresponding UniProt protein records.
UniProt	Provides protein sequence and functional annotation information.
Gene Ontology	Provides standardized functional labels used as prediction targets.

Table 1: Main data sources used in the DeepFRI dataset.

4.2 How the Data Were Collected

The DeepFRI authors did not create new proteins or determine new structures. They used protein structures that researchers had already deposited in PDB. For each selected protein chain, they identified the corresponding UniProt record using SIFTS mappings.

SIFTS, which stands for Structure Integration with Function, Taxonomy and Sequences, is a mapping resource that connects PDB protein chains with UniProt protein entries. This connection allowed the DeepFRI authors to associate structural data with functional annotations.

After identifying the corresponding UniProt records, the authors retrieved functional annotations, particularly Gene Ontology terms. These annotations were then linked with the protein sequence and structure-derived information to create training examples.

4.3 How the Data Were Processed

The raw PDB structures were processed into a format suitable for deep learning. Instead of using full three-dimensional coordinate files directly, the authors converted each protein structure into a contact map.

A contact map records which amino acids are close to one another in the three-dimensional structure. In the DeepFRI approach, the distance between C-alpha atoms was used to determine structural contacts. Two amino acids were considered connected when their C-alpha atoms were sufficiently close in space.

The protein sequence was also converted into numerical representations suitable for machine learning. The DeepFRI work used a pretrained protein language model to extract sequence-based features. Structural contacts were represented as a graph, where each amino acid was a node and nearby amino acids were connected by edges.

Finally, the sequence features, structural contact information, and functional labels were stored together in TensorFlow TFRecord format. The 19GB PDB–Gene Ontology package contains training and validation data in this processed format.

5. Proposed Methodology

The project will follow a structured research and development workflow. First, the downloaded dataset will be inspected and statistically analyzed. Then, baseline and structure-informed deep-learning models will be developed and evaluated.

The proposed workflow is shown below:

Dataset inspection → **statistical analysis** → **model development** → **model training** → **evaluation** → **software integration**

5.1 Dataset Analysis

Before training any model, the dataset will be analyzed to understand its characteristics. The analysis will include protein sequence lengths, amino-acid composition, frequency of functional labels, number of functions per protein, and co-occurrence of functions.

The analysis will also examine structural contact information, such as the number of contacts per protein and the distribution of contact-map sizes. This will help identify data imbalance, repeated proteins, and functions with insufficient examples.

5.2 Model Development

The first model will be a sequence-based baseline model. It will use protein sequence information to predict molecular-function labels. This model will provide a reference point for evaluating the benefit of structural information.

A second model will use both sequence information and structure-derived contact maps. The structural information will be represented as a graph, and a graph neural network will learn relationships between amino acids that are close in three-dimensional space.

The project will compare the performance of these models under the same dataset and evaluation conditions. This comparison will help determine whether structural information improves protein function prediction.

5.3 Model Training and Evaluation

The dataset will be divided into training, validation, and test sets. The training set will be used to learn model parameters, the validation set will be used to tune the model, and the test set will be used for final evaluation.

Because a protein can have multiple functional labels, evaluation will use multi-label classification metrics such as precision, recall, F1-score, F-max, and area under the precision-recall curve. These metrics are commonly used in protein function prediction research.

The project will also analyze performance across common and rare functional labels, protein sequence lengths, and different levels of structural information.

6. Proposed Software System

Alongside the research work, a software system will be developed to demonstrate the practical use of the trained model. The system will allow users to view protein information, submit a protein sequence, and receive predicted molecular functions.

Component	Technology	Purpose
Machine learning	Python, TensorFlow/PyTorch	Dataset processing, statistical analysis, model training, and prediction
Backend	Python, FastAPI	Connects the trained model, database, and frontend
Database	MySQL	Stores protein records, annotations, prediction history, and experiment metad
Frontend	React or HTML/CSS/JavaScript	Provides an interface for users to explore data and view predictions

Table 2: Proposed software components.

7. Expected Outcome

The expected outcome is a research-based protein function prediction system supported by experimental analysis. The project is expected to produce a trained deep-learning model, a comparative evaluation of sequence-based and structure-informed approaches, statistical insights into the dataset, and a working software prototype.

The final deliverables will include the source code, processed analysis results, trained model artifacts, database design, frontend interface, project report, and a research manuscript prepared for potential publication.

8. Limitations

The project will use an existing benchmark dataset rather than collecting new experimental protein structures. Functional annotations may be incomplete, and the absence of a label does not always mean that a protein cannot perform that function. In addition, model predictions are computational estimates and should not be treated as experimentally confirmed biological facts.

The project will initially focus on molecular-function prediction. Predicting protein structure from sequence, validating mutation effects experimentally, and covering all possible protein functions are outside the initial scope.

9. References

[1] V. Gligorijevi■ et al., “Structure-based protein function prediction using graph convolutional networks,” Nature Communications, vol. 12, art. 3168, 2021.

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